MANOHARI KULKARNI

POSTGRADUATE STUDENT - CANCER BIOLOGY



PROFILE

M.Sc. graduate in cancer biology seeking to pursue translational research. I am interested in how cancer cells survive and resist cell death, the focus of my research so far. A brief clinical exposure sharpened my goal and keeps my research interests grounded in patient needs. I enjoy making science clear and accessible to everyone. I aim to build an independent research career that brings discoveries to patients and improves lives.

EDUCATION

Aug 2024 – May 2026

Nitte University Centre for Science Education and Research (NUCSER), Mangaluru, Karnataka, India.

M.Sc in Cancer Biology

CGPA (up to 3rd semester): 8.07/10 Coursework focused on cancer genetics, cancer pathology, cell & molecular biology, immunology, and translational oncology with integrated rigorous laboratory training.

2021-2024

Dayananda Sagar University, Bengaluru, Karnataka, India.

B.Sc. in Biotechnology, Biochemistry and Genetics

CGPA: 7.07/10

Coursework built a strong foundation in biotechnology, biochemistry, genetics, microbiology, and core experimental biology techniques

2019-2021

BASE PU College, Bengaluru, Karnataka, India.

Class XII (Physics, Chemistry, Mathematics and Biology)

Score: 85.6%

RESEARCH PROJECTS

August,2025 – May,2026

Department of Molecular Genetics and Cancer, NUCSER

Role of Compensatory SUMOylation Mechanisms in NSMCE2-Deficient Breast Cancer Cells

- Investigating how NSMCE2 deficiency in triple-negative breast cancer may trigger compensatory SUMOylation programs that support survival during chemotherapeutic stress; mainly focused on PARP inhibitors (Olaparib)
- Focusing on links between SUMOylation-driven DNA repair adaptation, replication-stress tolerance, and chemoresistance to identify potential therapeutic targets.
- Built a Python tool to automate wound scratch-assay quantification for thesis data analysis.

January,2024 - June,2024

SynTech Bio Solutions Lab, Dayananda Sagar University

Rutin-Encapsulated PEG-PLGA nanoparticles for modulating autophagy and metabolism in Cancer

- Formulated and characterized Rutin-loaded PEG-PLGA nanoparticles to improve compound stability and bioavailability for targeted therapeutic applications.
- Integrated molecular docking with biological rationale to evaluate interactions with key autophagy/metabolic regulators and compare with established inhibitors

CONTACTS

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PUBLICATION

Book Chapter : **The Role of Epigenetics in Cancer Progression, Cancer Targets: Novel Therapies and Emerging Research Directions (Part 1)**

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CONFERENCES, WORKSHOPS & CERTIFICATES

- International Conference on Cancer Epigenetics, MAHE
- Cell Therapy Conclave, Manipal Centre for Biotherapeutics Research, Kasturba Medical College, MAHE, Manipal
- Symposium on Non-Invasive Imaging & Molecular Techniques, NUCSER
- Hacking the Gene Code (siRNA Design and Optimisation), Manipal Institute of Technology, Manipal
- International Hands-on Workshop: Utilizing microbiome and genomic resources for understanding and mitigating antimicrobial resistance in one health context (FAO Reference Centre, Nitte DU)

EXPERIENCE

June 2026 – July 2026

Orthopaedic Oncology & Sarcoma Care Team, HCG Cancer Centre, Bengaluru

Research Intern, The Purple Spark (medtech arm of The Yellow Ribbon)

- Contributed to clinical data collection for a study on 10+ year sarcoma survivors, recording details such as sarcoma subtype, treatment regimen and the multidisciplinary oncology team involved.
- Gained exposure to multidisciplinary decision-making in sarcoma diagnosis and limb-salvage surgery.

Aug 2026 - Present

Tapovan School, Bengaluru

Chemistry Teacher (Part-time)

- Teach ICSE/ISC chemistry (grades 7-12), pairing conceptual clarity with hands-on laboratory practicals that build analytical reasoning.

TECHNICAL SKILLS

Cell and Molecular Biology

- Mammalian Cell Culture & Maintenance
- qPCR
- Western blotting
- PCR
- Gel electrophoresis
- ELISA
- Transfection (siRNA/Plasmid)

Cancer and Nanobiotechnology

- In Vitro Cancer Models
- Nanoparticle Synthesis & Characterization
- Drug Delivery Systems
- Cytotoxicity & Phenotypic Assays

Data Analysis and Scientific Communication

- GraphPad Prism
- R (Basic packages)
- Python (Image analysis)

Computational and Bioinformatics Tools

- AutoDock
- Biovia Discovery Studio
- Chimera
- PyMOL
- BLAST
- Clustal Omega